

Catalysis **10** and Catalytic Reactors

It isn't that they can't see the solution. It is that they can't see the problem.

—G. K. Chesterton

Overview. The objectives of this chapter are to develop an understanding of catalysts, reaction mechanisms, and catalytic reactor design. Specifically, after reading this chapter one should be able to

- Define a catalyst and describe its properties.
- Describe the steps in a catalytic reaction and in chemical vapor deposition (CVD).
- Suggest a mechanism and apply the concept of a rate-limiting step to derive a rate law.
- Use nonlinear regression to determine the rate law and rate-law parameters that best fit the data.
- Use the rate law parameters to design PBRs and fluidized CSTRs.
- Describe the analogy between catalytic steps and CVD in microelectronic fabrication.
- Analyze catalyst decay and reactors that can be used to help offset decay.

The various sections of this chapter roughly correspond to these objectives.

10.1 Catalysts

Catalysts have been used by humankind for over 2000 years.¹ The first observed uses of catalysts were in the making of wine, cheese, and bread. It was found that it was always necessary to add small amounts of the previous

¹ S. T. Oyama and G. A. Somorjai, *J. Chem. Educ.*, 65, 765 (1986).

batch to make the current batch. However, it wasn't until 1835 that Berzelius began to tie together observations of earlier chemists by suggesting that small amounts of a foreign substance could greatly affect the course of chemical reactions. This mysterious force attributed to the substance was called *catalytic*. In 1894, Ostwald expanded Berzelius's explanation by stating that catalysts were substances that accelerate the rate of chemical reactions without being consumed during the reaction. During the 180 years since Berzelius's work, catalysts have come to play a major economic role in the world market. In the United States alone, sales of process catalysts will reach over \$20 billion by 2018, the major uses being in petroleum refining and in chemical production.

10.1.1 Definitions

A *catalyst* is a substance that affects the rate of a reaction but emerges from the process unchanged. A catalyst usually changes a reaction rate by promoting a different molecular path ("mechanism") for the reaction. For example, gaseous hydrogen and oxygen are virtually inert at room temperature, but react rapidly when exposed to platinum. The reaction coordinate (cf. Chapter 3) shown in Figure 10-1 is a measure of the progress along the reaction path as H_2 and O_2 approach each other and pass over the activation energy barrier to form H_2O . *Catalysis is the occurrence, study, and use of catalysts and catalytic processes.* Commercial chemical catalysts are immensely important. Approximately one-third of the material gross national product of the United States involves a catalytic process somewhere between raw material and finished product.² The development and use of catalysts is a major part of the constant search for new ways of increasing product yield and selectivity from chemical reactions. Because a catalyst makes it possible to obtain an end product by a different pathway with a lower energy barrier, it can affect both the yield and the selectivity.

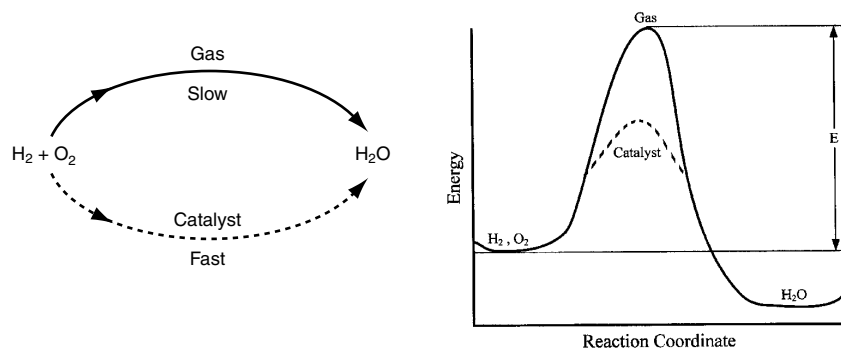


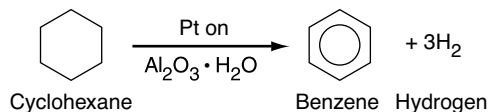
Figure 10-1 Different reaction paths.

Catalysts can accelerate the reaction rate but cannot change the equilibrium.

Normally when we talk about a catalyst, we mean one that speeds up a reaction, although strictly speaking, a catalyst can either accelerate or slow the formation of a particular product species. *A catalyst changes only the rate of a reaction; it does not affect the equilibrium.*

² V. Haensel and R. L. Burwell, Jr., *Sci. Am.*, 225(10), 46.

The 2007 Nobel Prize for Chemistry was awarded to Gerhard Ertl for his pioneering work on heterogeneous catalytic reactions. A *heterogeneous catalytic reaction* involves more than one phase; usually the catalyst is a solid and the reactants and products are in liquid or gaseous form. One example is the production of benzene, which is mostly manufactured today from the dehydrogenation of cyclohexane (obtained from the distillation of petroleum crude oil) using platinum-on-alumina as the catalyst:



The simple and complete separation of the fluid product mixture from the solid catalyst makes heterogeneous catalysis economically attractive, especially because many catalysts are quite valuable and their reuse is demanded.

A heterogeneous catalytic reaction occurs at or very near the fluid–solid interface. The principles that govern heterogeneous catalytic reactions can be applied to both catalytic and noncatalytic fluid–solid reactions. The two other types of heterogeneous reactions involve gas–liquid and gas–liquid–solid systems. Reactions between gases and liquids are usually mass-transfer limited.

10.1.2 Catalyst Properties

Ten grams of this catalyst possess more surface area than a U.S. football field

Catalyst types:

- Porous
- Molecular sieves
- Monolithic
- Supported
- Unsupported

Typical zeolite catalyst

High selectivity to para-xylene

Because a catalytic reaction occurs at the fluid–solid interface, a large interfacial area is almost always essential in attaining a significant reaction rate. In many catalysts, this area is provided by an inner porous structure (i.e., the solid contains many fine pores, and the surface of these pores supplies the area needed for the high rate of reaction), see Figures 10-4(b) and 10-9. The area possessed by some porous catalysis materials is surprisingly large. A typical silica-alumina cracking catalyst has a pore volume of 0.6 cm³/g and an average pore radius of 4 nm. The corresponding surface area is 300 m²/g of these *porous catalysts*. Examples include the Raney nickel used in the hydrogenation of vegetable and animal oils, platinum-on-alumina used in the reforming of petroleum naphthas to obtain higher octane ratings, and promoted iron used in ammonia synthesis. Sometimes pores are so small that they will admit small molecules but prevent large ones from entering. Materials with this type of pore are called *molecular sieves*, and they may be derived from natural substances such as certain clays and zeolites, or they may be totally synthetic, such as some crystalline aluminosilicates (see Figure 10-2). These sieves can form the basis for quite selective catalysts; the pores can control the residence time of various molecules near the catalytically active surface to a degree that essentially allows *only* the desired molecules to react. One example of the high selectivity of zeolite catalysts is the formation of para-xylene from toluene and methane shown in Figure 10-2(b).³ Here, benzene and toluene enter through the zeolite pore and react on the interior surface to form a mixture of ortho-, meta-, and para-xylenes. However, the size of the pore mouth is such that only

³ R. I. Masel, *Chemical Kinetics and Catalysis* (New York: Wiley Interscience, 2001), p. 741.

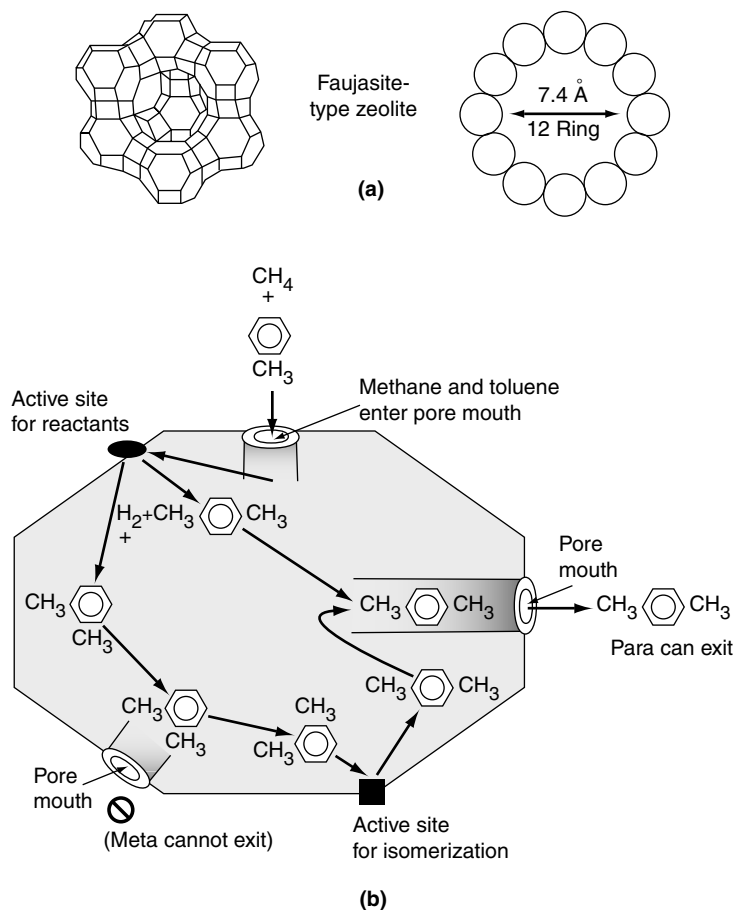


Figure 10-2 (a) Framework structures and (b) pore cross sections of two types of zeolites. (a) Faujasite-type zeolite has a three-dimensional channel system with pores at least 7.4 Å in diameter. A pore is formed by 12 oxygen atoms in a ring. (b) Schematic of reaction CH₄ and C₆H₅CH₃. (Note that the size of the pore mouth and the interior of the zeolite are not to scale.) [(a) from N. Y. Chen and T. F. Degnan, *Chem. Eng. Prog.*, 84(2), 33 (1988). Reproduced by permission of the American Institute of Chemical Engineers. Copyright © 1988 AIChE. All rights reserved.]

para-xylene can exit through the pore mouth, as meta- and ortho- xylene with their methyl group on the side cannot fit through the pore mouth. There are interior sites that can isomerize ortho-xylene and meta-xylene to para-xylene. Hence, we have a very high selectivity to form para-xylene.

In some cases a catalyst consists of minute particles of an active material dispersed over a less-active substance called a *support*. The active material is frequently a pure metal or metal alloy. Such catalysts are called *supported catalysts*, as distinguished from *unsupported catalysts*. Catalysts can also have small amounts of active ingredients added called *promoters*, which increase their activity. Examples of supported catalysts are the packed-bed catalytic converter in an automobile, the platinum-on-alumina catalyst used in petroleum reforming, and the vanadium pentoxide on silica used to oxidize sulfur dioxide in manufacturing sulfuric acid. On the other hand, platinum gauze for ammonia oxidation, promoted iron for ammonia synthesis, and the silica–alumina dehydrogenation catalyst used in butadiene manufacture typify unsupported catalysts.

10.1.3 Catalytic Gas-Solid Interactions

For the moment, let us focus our attention on gas-phase reactions catalyzed by solid surfaces. For a catalytic reaction to occur, at least one and frequently all of the reactants must become attached to the surface. This attachment is known as *adsorption* and takes place by two different processes: physical adsorption and chemisorption. *Physical adsorption* is similar to condensation. The process is exothermic, and the heat of adsorption is relatively small, being on the order of 1 to 15 kcal/mol. The forces of attraction between the gas molecules and the solid surface are weak. These van der Waals forces consist of interaction between permanent dipoles, between a permanent dipole and an induced dipole, and/or between neutral atoms and molecules. The amount of gas physically adsorbed decreases rapidly with increasing temperature, and above its critical temperature only very small amounts of a substance are physically adsorbed.

The type of adsorption that affects the rate of a chemical reaction is *chemisorption*. Here, the adsorbed atoms or molecules are held to the surface by valence forces of the same type as those that occur between bonded atoms in molecules. As a result, the electronic structure of the chemisorbed molecule is perturbed significantly, causing it to be extremely reactive. Interaction with the catalyst causes bonds of the adsorbed reactant to be stretched, making them easier to break.

Figure 10-3 shows the bonding from the adsorption of ethylene on a platinum surface to form chemisorbed ethylidyne. Like physical adsorption, chemisorption is an exothermic process, but the heats of adsorption are

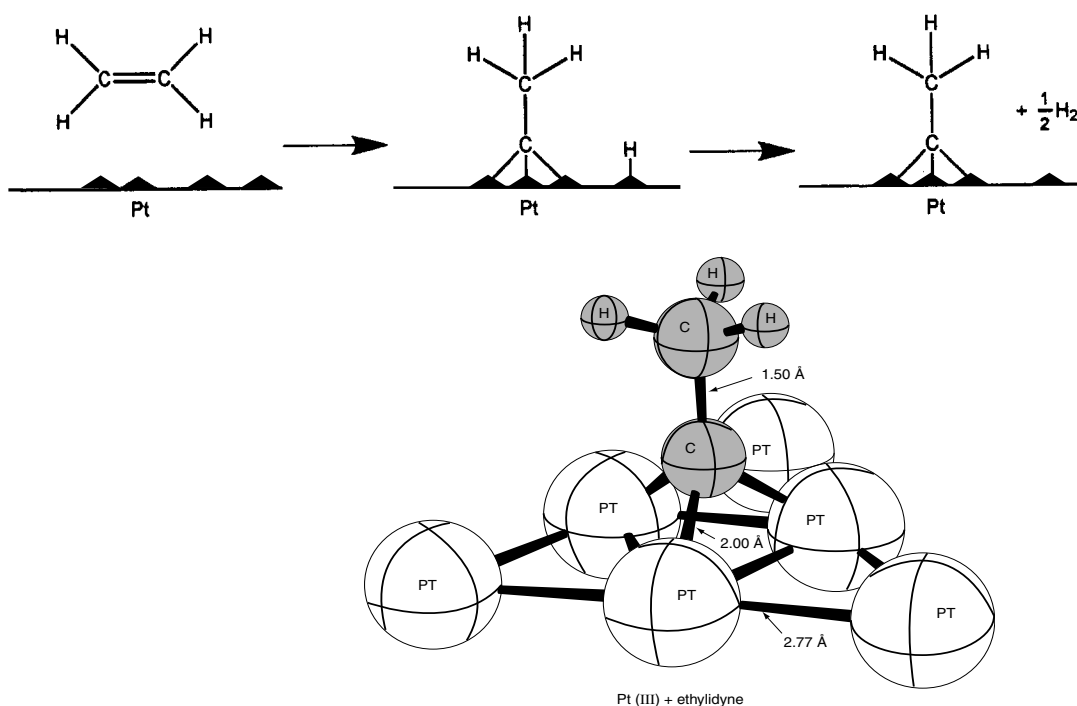


Figure 10-3 Ethylidyne chemisorbed on platinum. (Adapted from G. A. Somorjai, *Introduction to Surface Chemistry and Catalysis*. © 1994 John Wiley & Sons, Inc. Reprinted by permission of John Wiley & Sons, Inc. All rights reserved.)

Chemisorption on active sites is what catalyzes the reaction.

generally of the same magnitude as the heat of a chemical reaction (i.e., 40 to 400 kJ/mol). If a catalytic reaction involves chemisorption, it must be carried out within the temperature range where chemisorption of the reactants is appreciable.

In a landmark contribution to catalytic theory, Taylor suggested that a reaction is not catalyzed over the entire solid surface but only at certain *active sites* or *centers*.⁴ He visualized these sites as unsaturated atoms in the solids that resulted from surface irregularities, dislocations, edges of crystals, and cracks along grain boundaries. Other investigators have taken exception to this definition, pointing out that other properties of the solid surface are also important. The active sites can also be thought of as places where highly reactive intermediates (i.e., chemisorbed species) are stabilized long enough to react. This stabilization of a reactive intermediate is key in the design of any catalyst. Consequently, for our purposes we will define an *active site* as a *point on the catalyst surface that can form strong chemical bonds with an adsorbed atom or molecule*.

One parameter used to quantify the activity of a catalyst is the *turnover frequency (TOF)*, f . It is the number of molecules reacting per active site per second at the conditions of the experiment. When a metal catalyst such as platinum is deposited on a support, the metal atoms are considered active sites. The TOFs for a number of reactions are shown in *Professional Reference Shelf R10.1*. The *dispersion*, D , of the catalyst is the fraction of the metal atoms deposited on a catalyst that are *on* the surface.

10.1.4 Classification of Catalysts

One common way to classify catalysts is in terms of the type of reaction they catalyze.

Table 10-1 gives a list of representative reactions and their corresponding catalysts. Further discussion of each of these reaction classes and the materials that catalyze them can be found on the CRE Web site's *Professional Reference Shelf R10.1*.

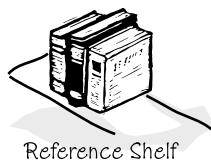


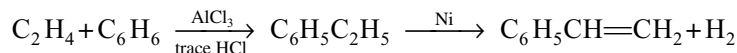
TABLE 10-1 TYPES OF REACTIONS AND REPRESENTATIVE CATALYSTS

<i>Reaction</i>	<i>Catalysts</i>
1. Halogenation–dehalogenation	CuCl ₂ , AgCl, Pd
2. Hydration–dehydration	Al ₂ O ₃ , MgO
3. Alkylation–dealkylation	AlCl ₃ , Pd, Zeolites
4. Hydrogenation–dehydrogenation	Co, Pt, Cr ₂ O ₃ , Ni
5. Oxidation	Cu, Ag, Ni, V ₂ O ₅
6. Isomerization	AlCl ₃ , Pt/Al ₂ O ₃ , Zeolites

If, for example, we were to form styrene from an equimolar mixture of ethylene and benzene, we could first carry out an alkylation reaction to form

⁴ H. S. Taylor, *Proc. R. Soc. London*, A108, 105 (1928).

ethyl benzene, which is then dehydrogenated to form styrene. We need both an alkylation catalyst and a dehydrogenation catalyst:



10.2 Steps in a Catalytic Reaction

A photograph of different types and sizes of catalysts is shown in Figure 10-4a. A schematic diagram of a tubular reactor packed with catalytic pellets is shown in Figure 10-4b. The overall process by which heterogeneous catalytic reactions proceed can be broken down into the sequence of individual steps shown in Table 10-2 and pictured in Figure 10-5 for an isomerization reaction.



Figure 10-4a Catalyst particles of different shapes (spheres, cylinders) and sizes (0.1 cm to 1 cm). (Photo courtesy of the Engelhard Corporation.)

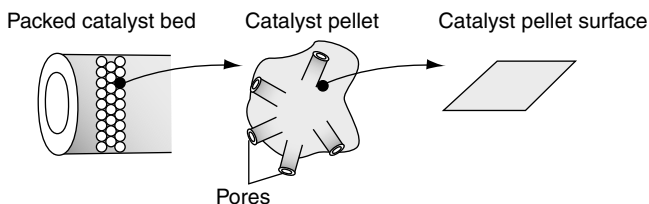


Figure 10-4b Catalytic packed-bed reactor—schematic.

Each step in Table 10-2 is shown schematically in Figure 10-5.

A reaction takes place *on* the surface, but the species involved in the reaction must get *to* and *from* the surface.

The overall rate of reaction is limited by the rate of the slowest step in the sequence. When the diffusion steps (1, 2, 6, and 7 in Table 10-2) are very fast compared with the surface reaction-rate steps (3, 4, and 5), the concentrations in the immediate vicinity of the active sites are indistinguishable from those in the bulk fluid. In this situation, the transport or diffusion steps do not affect the overall rate of the reaction. In other situations, if the reaction steps are very fast compared with the diffusion steps, mass transport does affect the reaction rate. In systems where diffusion from the bulk gas or liquid to the external catalyst surface or to the mouths of catalyst pores affects the rate, i.e., steps 1 and 7, changing the flow conditions past the catalyst should change the overall reaction rate (see Chapter 14). Once inside the porous catalysts, on the other hand, diffusion within the catalyst pores, i.e., steps 2 and 6, may limit the rate of reaction and, as a result, the overall rate will be unaffected by external flow conditions even though diffusion affects the overall reaction rate (see Chapter 15).

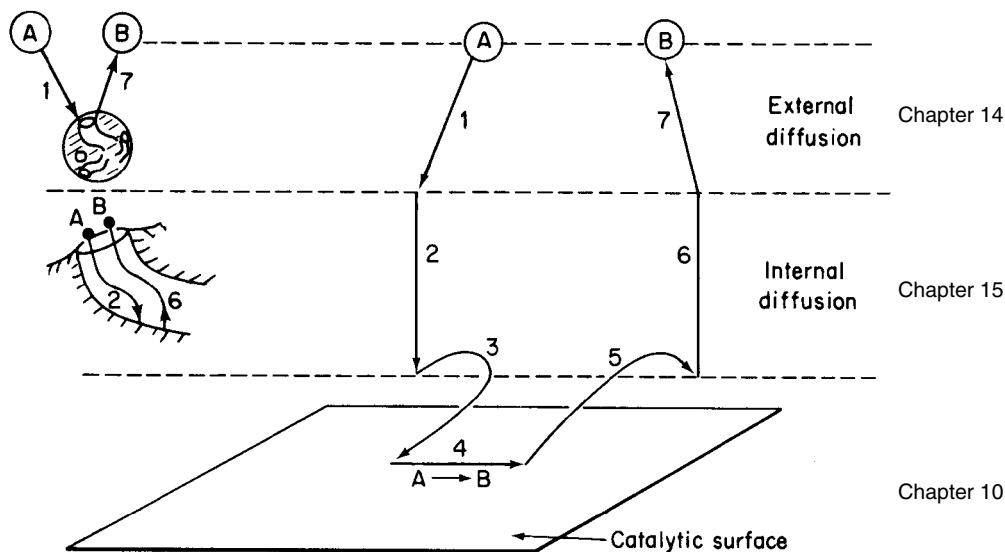


Figure 10-5 Steps in a heterogeneous catalytic reaction.

TABLE 10-2 STEPS IN A CATALYTIC REACTION

1. Mass transfer (diffusion) of the reactant(s) (e.g., species A) from the bulk fluid to the external surface of the catalyst pellet
2. Diffusion of the reactant from the pore mouth through the catalyst pores to the immediate vicinity of the internal catalytic surface
3. Adsorption of reactant A onto the catalyst surface
4. Reaction on the surface of the catalyst (e.g., $A \longrightarrow B$)
5. Desorption of the products (e.g., B) from the surface
6. Diffusion of the products from the interior of the pellet to the pore mouth at the external surface
7. Mass transfer of the products from the external pellet surface to the bulk fluid

There are many variations of the situation described in Table 10-2. Sometimes, of course, two reactants are necessary for a reaction to occur, and both of these may undergo the steps listed above. Other reactions between two substances may have only one of them adsorbed.

With this introduction, we are ready to treat individually the steps involved in catalytic reactions. In this chapter, only steps 3, 4, and 5—i.e., adsorption, surface reaction, and desorption—are considered as we assume that the diffusion steps (1, 2, 6, and 7) are very fast so that the overall reaction rate is not affected by mass transfer in any fashion. Further treatment of the effects involving diffusion limitations is provided in Chapters 14 and 15.

Where Are We Heading?[†] As we saw in Chapter 7, one of the tasks of a chemical reaction engineer is to analyze rate data and to develop a rate law that can be used in reactor design. Rate laws in heterogeneous catalysis seldom

In this chapter we focus on:
3. Adsorption
4. Surface reaction
5. Desorption

[†] “If you don’t know where you are going, you’ll probably wind up someplace else.”
Yogi Berra

An algorithm

follow power-law models and hence are inherently more difficult to formulate from the data. To develop an in-depth understanding and insight as to how the rate laws are formed from heterogeneous catalytic data, we are going to proceed in somewhat of a reverse manner than what is normally done in industry when one is asked to develop a rate law. That is, we will postulate catalytic mechanisms and *then* derive rate laws for the various mechanisms. The mechanism will typically have an adsorption step, a surface reaction step, and a desorption step, one of which is usually rate-limiting. Suggesting mechanisms and rate-limiting steps is not the first thing we normally do when presented with data. However, by deriving equations for different mechanisms, we will observe the various forms of the rate law one can have in heterogeneous catalysis. Knowing the different forms that catalytic rate equations can take, it will be easier to view the trends in the data and deduce the appropriate rate law. This deduction is usually what is done first in industry before a mechanism is proposed. Knowing the form of the rate law, one can then numerically evaluate the rate-law parameters and postulate a reaction mechanism and rate-limiting step that are consistent with the rate data. Finally, we use the rate law to design catalytic reactors. This procedure is shown in Figure 10-6. The dashed lines represent feedback to obtain new data in specific regions (e.g., concentrations, temperature) to evaluate the rate-law parameters more precisely or to differentiate between reaction mechanisms.

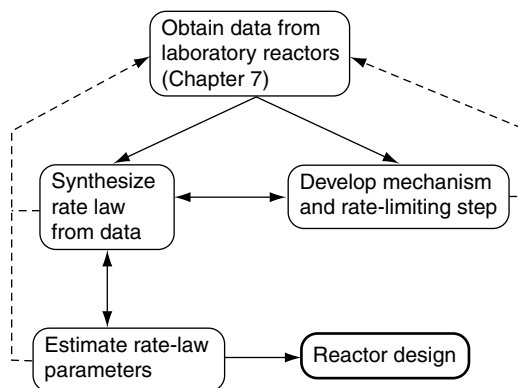


Figure 10-6 Collecting information for catalytic reactor design.

Chance Card:
Do not pass go
proceed directly to
Section 10.2.3.



We will discuss each of the steps shown in Figure 10-5 and Table 10-2. As mentioned earlier, this chapter focuses on Steps 3, 4, and 5 (the adsorption, surface reaction and desorption steps) by assuming that Steps 1, 2, 6, and 7 are very rapid. Consequently, to understand when this assumption is valid, we shall give a quick overview of Steps 1, 2, 6, and 7. These steps involve diffusion of the reactants to and within the catalyst pellet. While these diffusion steps are covered in detail in Chapters 14 and 15, it is worthwhile to give a brief description of these two mass-transfer steps to better understand the entire sequence of steps. If you have had the core course in *Mass Transfer* or *Transport Phenomena* you can skip Sections 10.2.1 and 10.2.2, and go directly to Section 10.2.3.

10.2.1 Step 1 Overview: Diffusion from the Bulk to the External Surface of the Catalyst

External and internal mass transfer in catalysis are covered in detail in Chapters 14 and 15.

For the moment let's assume that the transport of A from the bulk fluid to the external surface of the catalyst is the slowest step in the sequence. We lump all the resistance to transfer from the bulk fluid to the surface in the mass-transfer boundary layer surrounding the pellet. In this step the reactant A, which is at a bulk concentration C_{Ab} must travel (diffuse) through the boundary layer of thickness δ to the external surface of the pellet where the concentration is C_{As} , as shown in Figure 10-7. The rate of transfer (and hence rate of reaction, $-r'_A$) for this slowest step is

$$\text{Rate} = k_C (C_{Ab} - C_{As})$$

where the mass-transfer coefficient, k_C , is a function of the hydrodynamic conditions, namely the fluid velocity, U , and the particle diameter, D_p .

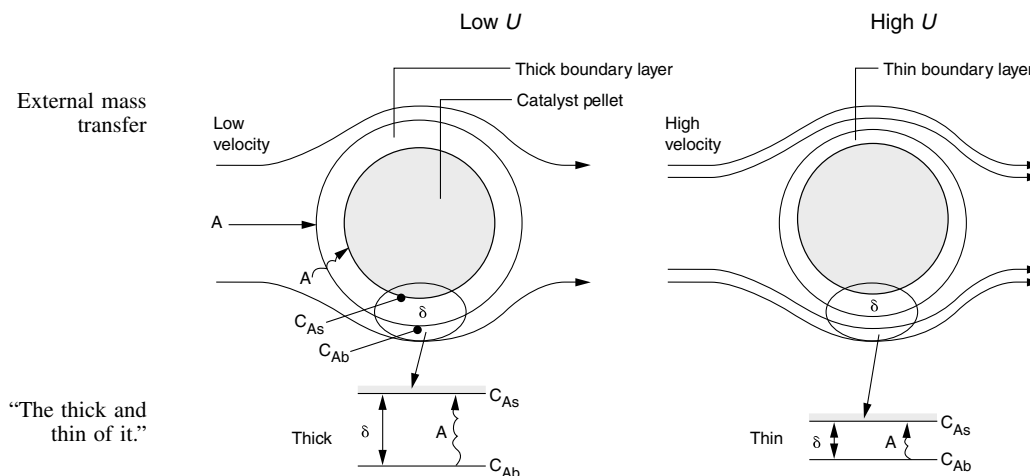


Figure 10-7 Diffusion through the external boundary layer (also see Figure 14-3).

As we see (Chapter 14), the mass-transfer coefficient is inversely proportional to the boundary layer thickness, δ , and directly proportional to the diffusion coefficient (i.e., the diffusivity D_{AB}).

$$k_C = \frac{D_{AB}}{\delta}$$

At low velocities of fluid flow over the pellet, the boundary layer across which A and B must diffuse is thick, and it takes a long time for A to travel to the surface, resulting in a small mass-transfer coefficient k_C . As a result, mass transfer across the boundary layer is slow and limits the rate of the overall reaction. As the velocity over the pellet is increased, the boundary layer becomes thinner and the mass-transfer rate is increased. At very high velocities, the boundary layer thickness, δ , is so small it no longer offers any resistance to the diffusion across

the boundary layer. As a result, external mass transfer no longer limits the rate of reaction. This external resistance also decreases as the particle size is decreased. As the fluid velocity increases and/or the particle diameter decreases, the mass-transfer coefficient increases until a plateau is reached, as shown in Figure 10-8. On this plateau, $C_{Ab} \approx C_{As}$, and one of the other steps in the sequence, i.e., Steps 2 through 6, is the slowest step and limits the overall reaction rate. Further details on external mass transfer are discussed in Chapter 14.

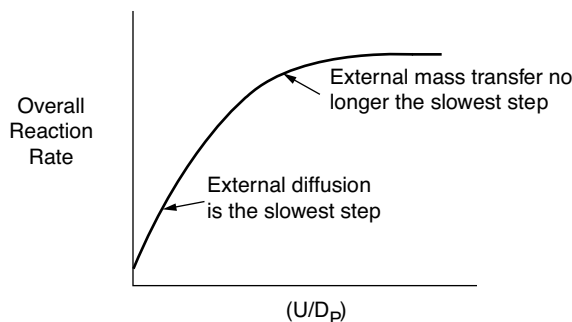


Figure 10-8 Effect of particle size and fluid velocity on the overall reaction rate.

10.2.2 Step 2 Overview: Internal Diffusion

Now consider that we are operating at a fluid velocity where external diffusion is no longer the rate-limiting step and that internal diffusion is the slowest step. In Step 2 the reactant A diffuses from the external pellet surface at a concentration C_{As} into the pellet interior, where the concentration is C_A . As A diffuses into the interior of the pellet, it reacts with catalyst deposited on the sides of the catalyst pellet's pore walls.

For large pellets, it takes a long time for the reactant A to diffuse into the interior, compared to the time that it takes for the reaction to occur on the interior pore surface. Under these circumstances, the reactant is only consumed near the exterior surface of the pellet and the catalyst near the center of the pellet is wasted catalyst. On the other hand, for very small pellets it takes very little time to diffuse into and out of the pellet interior and, as a result, internal diffusion no longer limits the rate of reaction. When internal mass transfer no longer limits the rate of reaction, the rate law can be expressed as

$$\text{Rate} = k_r C_{As}$$

where C_{As} is the concentration at the external pellet surface and k_r is an overall rate constant, which is a function of particle size. The overall rate constant, k_r , increases as the pellet diameter decreases. In Chapter 15, we show that Figure 15-5 can be combined with Equation (15-34) to arrive at the plot of k_r as a function of D_p , shown in Figure 10-9(b).

We see in Figure 10-9 that at small particle sizes, internal diffusion is no longer the slow step and that the surface reaction sequence of adsorption, surface reaction, and desorption (Steps 3, 4, and 5 in Figure 10-5) limit the overall rate

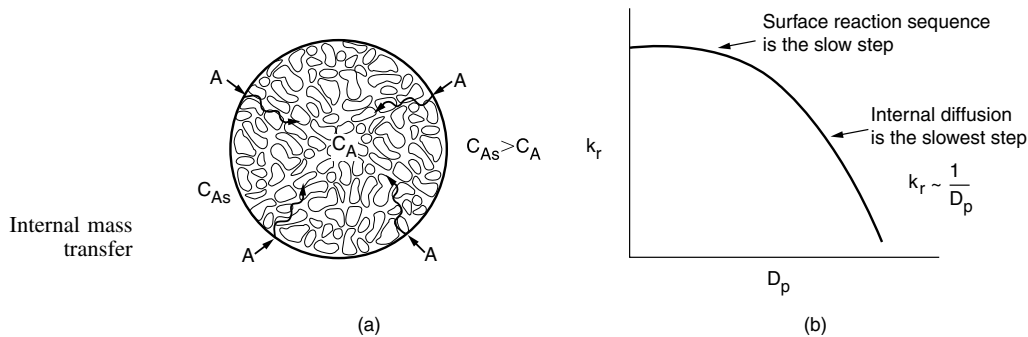


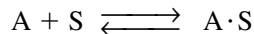
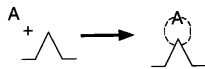
Figure 10-9 Effect of particle size on the overall reaction-rate constant. (a) Branching of a single pore with deposited metal; (b) decrease in rate constant with increasing particle diameter. (See the CRE Web site, Chapter 12.)

of reaction. Consider now one more important point about internal diffusion and surface reaction. These steps (2 through 6) are **not at all** affected by flow conditions external to the pellet.

In the material that follows, we are going to choose our pellet size and external fluid velocity such that neither external diffusion nor internal diffusion is limiting as discussed in Chapters 14 and 15. Instead, we assume that either Step 3 (adsorption), Step 4 (surface reaction), or Step 5 (desorption), or a combination of these steps, limits the overall rate of reaction.

10.2.3 Adsorption Isotherms

Because chemisorption is usually a necessary part of a catalytic process, we shall discuss it before treating catalytic reaction rates. The letter S will represent an active site; alone, it will denote a vacant site, with no atom, molecule, or complex adsorbed on it. The combination of S with another letter (e.g., A·S) will mean that one unit of species A will be chemically adsorbed on the site S. Species A can be an atom, molecule, or some other atomic combination, depending on the circumstances. Consequently, the adsorption of A on a site S is represented by



The total molar concentration of active sites per unit mass of catalyst is equal to the number of active sites per unit mass divided by Avogadro's number and will be labeled C_t (mol/g-cat). The molar concentration of vacant sites, C_v (mol/g-cat), is the number of vacant sites per unit mass of catalyst divided by Avogadro's number. In the absence of catalyst deactivation, we assume that the total concentration of active sites, C_t , remains constant. Some further definitions include

P_i = partial pressure of species i in the gas phase, (atm or kPa)

$C_{i \cdot S}$ = surface concentration of sites occupied by species i , (mol/g-cat)

A conceptual model depicting species A and B adsorbed on two different sites is shown in Figure 10-10.



Figure 10-10 Vacant and occupied sites.

For the system shown in Figure 10-10, the total concentration of sites is

Site balance

$$C_t = C_v + C_{A \cdot S} + C_{B \cdot S} \quad (10-1)$$

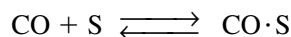
This equation is referred to as a *site balance*. A typical value for the total concentration of sites could be the order of 10^{22} sites/g-cat.

Now consider the adsorption of a nonreacting gas onto the surface of a catalyst. Adsorption data are frequently reported in the form of adsorption *isotherms*. Isotherms portray the amount of a gas adsorbed on a solid at different pressures at a given temperature.

Postulate models; then see which one(s) fit(s) the data.

First, an adsorption mechanism is proposed, and then the isotherm (see Figure 10-11, page 413) obtained from the mechanism is compared with the experimental data. If the isotherm predicted by the model agrees with the experimental data, the model may reasonably describe what is occurring physically in the real system. If the predicted curve does not agree with the experimental data, the model fails to match the physical situation in at least one important characteristic and perhaps more.

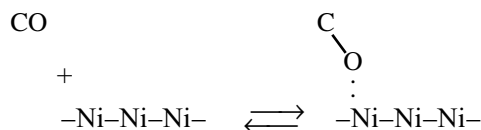
We will consider two types of adsorption: molecular adsorption and dissociative adsorption. To illustrate the difference between molecular adsorption and dissociative adsorption, we will postulate two models for the adsorption of carbon monoxide on metal surfaces. In the molecular adsorption model, CO is adsorbed as molecules, CO,



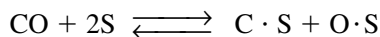
as is the case on nickel

Two models:

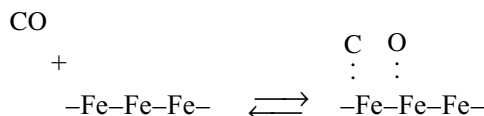
1. Adsorption as CO
2. Adsorption as C and O



In the dissociative adsorption model, carbon monoxide is adsorbed as oxygen and carbon atoms instead of molecular CO



as is the case on iron⁵



⁵ R. I. Masel, *Principles of Adsorption and Reaction on Solid Surfaces* (New York: Wiley, 1996).

The former is called *molecular* or *nondissociated adsorption* (e.g., CO) and the latter is called *dissociative adsorption* (e.g., C and O). Whether a molecule adsorbs nondissociatively or dissociatively depends on the surface.

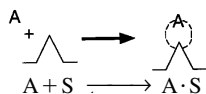
The adsorption of carbon monoxide molecules will be considered first. Because the carbon monoxide does not react further after being adsorbed, we need only to consider the adsorption process:

Molecular Adsorption



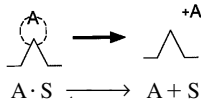
In obtaining a rate law for the rate of adsorption, the reaction in Equation (10-2) can be treated as an *elementary reaction*. The rate of attachment of the carbon monoxide molecules to the active site on the surface is proportional to the number of collisions that these molecules make with a surface active site per second. In other words, a specific fraction of the molecules that strike the surface become adsorbed. The collision rate is, in turn, directly proportional to the carbon monoxide partial pressure, P_{CO} . Because carbon monoxide molecules adsorb only on vacant sites and not on sites already occupied by other carbon monoxide molecules, the rate of attachment is also directly proportional to the concentration of vacant sites, C_v . Combining these two facts means that the rate of attachment of carbon monoxide molecules to the surface is directly proportional to the product of the partial pressure of CO and the concentration of vacant sites; that is,

$$P_{\text{CO}} = C_{\text{CO}}RT$$



$$\text{Rate of attachment} = k_A P_{\text{CO}} C_v$$

The rate of detachment of molecules from the surface can be a first-order process; that is, the detachment of carbon monoxide molecules from the surface is usually directly proportional to the concentration of sites occupied by the adsorbed molecules (e.g., $C_{\text{CO} \cdot \text{S}}$):



$$\text{Rate of detachment} = k_{-A} C_{\text{CO} \cdot \text{S}}$$

The net rate of adsorption is equal to the rate of molecular attachment to the surface minus the rate of detachment from the surface. If k_A and k_{-A} are the constants of proportionality for the attachment and detachment processes, then

$$r_{\text{AD}} = k_A P_{\text{CO}} C_v - k_{-A} C_{\text{CO} \cdot \text{S}} \quad (10-3)$$

The ratio $K_A = k_A/k_{-A}$ is the *adsorption equilibrium constant*. Using K_A to rearrange Equation (10-3) gives

$$\begin{aligned} \text{A} + \text{S} &\xrightleftharpoons{\text{Adsorption}} \text{A} \cdot \text{S} \\ r_{\text{AD}} &= k_A \left(P_A C_v - \frac{C_{\text{A} \cdot \text{S}}}{K_A} \right) \end{aligned}$$

$$r_{\text{AD}} = k_A \left(P_{\text{CO}} C_v - \frac{C_{\text{CO} \cdot \text{S}}}{K_A} \right) \quad (10-4)$$

The adsorption rate constant, k_A , for molecular adsorption is virtually independent of temperature, while the desorption constant, k_{-A} , increases exponentially with increasing temperature. Consequently, the *equilibrium adsorption constant* K_A *decreases exponentially with increasing temperature*.

Because carbon monoxide is the only material adsorbed on the catalyst, the site balance gives

$$C_t = C_v + C_{\text{CO} \cdot \text{S}} \quad (10-5)$$

$$-r'_A = r_{AD} = \left(\frac{\text{mol}}{\text{g-cat} \cdot \text{s}} \right)$$

$$k_A = \left(\frac{1}{\text{atm} \cdot \text{s}} \right)$$

$$P_A = (\text{atm})$$

$$C_v = \left(\frac{\text{mol}}{\text{g-cat}} \right)$$

$$K_A = \left(\frac{1}{\text{atm}} \right)$$

$$C_{A \cdot S} = \left(\frac{\text{mol}}{\text{g-cat}} \right)$$

At equilibrium, the net rate of adsorption equals zero, i.e., $r_{AD} \equiv 0$. Setting the left-hand side of Equation (10-4) equal to zero and solving for the concentration of CO adsorbed on the surface, we get

$$C_{\text{CO} \cdot S} = K_A C_v P_{\text{CO}} \quad (10-6)$$

Using Equation (10-5) to give C_v in terms of $C_{\text{CO} \cdot S}$ and the total number of sites C_t , we can solve for the equilibrium value of $C_{\text{CO} \cdot S}$ in terms of constants and the pressure of carbon monoxide

$$C_{\text{CO} \cdot S} = K_A C_v P_{\text{CO}} = K_A P_{\text{CO}} (C_t - C_{\text{CO} \cdot S})$$

Rearranging gives us

$$C_{\text{CO} \cdot S} = \frac{K_A P_{\text{CO}} C_t}{1 + K_A P_{\text{CO}}} \quad (10-7)$$

This equation thus gives the equilibrium concentration of carbon monoxide adsorbed on the surface, $C_{\text{CO} \cdot S}$, as a function of the partial pressure of carbon monoxide, and is an equation for the adsorption isotherm. This particular type of isotherm equation is called a *Langmuir isotherm*.⁶ **Figure 10-11(a)** shows the Langmuir isotherm for the amount of CO adsorbed per unit mass of catalyst as a function of the partial pressure of CO. For the case of dissociative adsorption, Equation (10-11), **Figure 10-11(b)**, shows the concentration of the atoms C and O adsorbed per unit mass of catalyst.

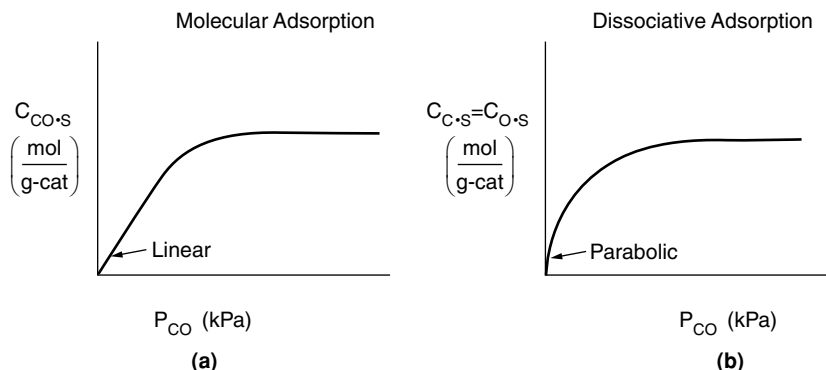


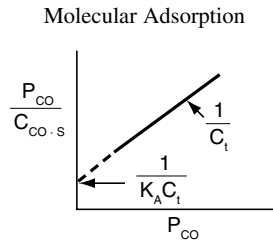
Figure 10-11 Langmuir isotherms for (a) molecular adsorption (b) dissociative adsorption of CO.

One method of checking whether a model (e.g., molecular adsorption versus dissociative adsorption) predicts the behavior of the experimental data is to linearize the model's equation and then plot the indicated variables against one another. For example, the molecular adsorption isotherm, Equation (10-7), may be arranged in the form

$$\frac{P_{\text{CO}}}{C_{\text{CO} \cdot S}} = \frac{1}{K_A C_t} + \frac{P_{\text{CO}}}{C_t} \quad (10-8)$$

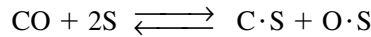
⁶ Named after Irving Langmuir (1881–1957), who first proposed it. He received the Nobel Prize in 1932 for his discoveries in surface chemistry.

and the linearity of a plot of $P_{\text{CO}}/C_{\text{CO}\cdot\text{S}}$ as a function of P_{CO} will determine if the data conform to molecular adsorption, i.e., a Langmuir single-site isotherm.



Next, we derive the isotherm for carbon monoxide disassociating into separate atoms as it adsorbs on the surface, i.e.,

Dissociative
adsorption



When the carbon monoxide molecule dissociates upon adsorption, it is referred to as the *dissociative adsorption* of carbon monoxide. As in the case of molecular adsorption, the rate of adsorption is proportional to the pressure of carbon monoxide in the system because this rate is governed by the number of gaseous collisions with the surface. For a molecule to dissociate as it adsorbs, however, two adjacent vacant active sites are required, rather than the single site needed when a substance adsorbs in its molecular form. The probability of two vacant sites occurring adjacent to one another is proportional to the square of the concentration of vacant sites. These two observations mean that the rate of adsorption is proportional to the product of the carbon monoxide partial pressure and the square of the vacant-site concentration, $P_{\text{CO}}C_v^2$.

For desorption to occur, two occupied sites must be adjacent, meaning that the rate of desorption is proportional to the product of the occupied-site concentration, $(\text{C}\cdot\text{S}) \times (\text{O}\cdot\text{S})$. The net rate of adsorption can then be expressed as

$$r_{\text{AD}} = k_{\text{A}}P_{\text{CO}}C_v^2 - k_{-\text{A}}C_{\text{O}\cdot\text{S}}C_{\text{C}\cdot\text{S}} \quad (10-9)$$

Factoring out k_{A} , the equation for *dissociative adsorption* is

Rate of dissociative
adsorption

$$r_{\text{AD}} = k_{\text{A}} \left(P_{\text{CO}}C_v^2 - \frac{C_{\text{C}\cdot\text{S}}C_{\text{O}\cdot\text{S}}}{K_{\text{A}}} \right)$$

where

$$K_{\text{A}} = \frac{k_{\text{A}}}{k_{-\text{A}}}$$

For dissociative adsorption, both k_{A} and $k_{-\text{A}}$ increase exponentially with increasing temperature, while the *adsorption equilibrium constant* K_{A} decreases with increasing temperature.

At equilibrium, $r_{\text{AD}} \equiv 0$, and

$$k_{\text{A}}P_{\text{CO}}C_v^2 = k_{-\text{A}}C_{\text{C}\cdot\text{S}}C_{\text{O}\cdot\text{S}}$$

For $C_{\text{C}\cdot\text{S}} = C_{\text{O}\cdot\text{S}}$

$$(K_{\text{A}}P_{\text{CO}})^{1/2}C_v = C_{\text{O}\cdot\text{S}} \quad (10-10)$$

Substituting for $C_{\text{C}\cdot\text{S}}$ and $C_{\text{O}\cdot\text{S}}$ in a site balance equation (10-1),

$$\begin{aligned}
 C_t &= C_v + C_{O \cdot S} + C_{C \cdot S} \\
 \text{Site balance:} \quad &= C_v + (K_{CO}P_{CO})^{1/2}C_v + (K_{CO}P_{CO})^{1/2}C_v = C_v(1 + 2(K_{CO}P_{CO})^{1/2})
 \end{aligned}$$

Solving for C_v

$$C_v = C_t / (1 + 2(K_{CO}P_{CO})^{1/2})$$

This value may be substituted into Equation (10-10) to give an expression that can be solved for the equilibrium value of $C_{O \cdot S}$. The resulting equation for the isotherm shown in Figure 10-11(b) is

Langmuir isotherm
for carbon monoxide
dissociative
adsorption as
atomic carbon and
oxygen

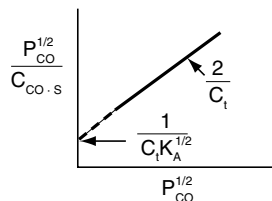
$$C_{O \cdot S} = \frac{(K_A P_{CO})^{1/2} C_t}{1 + 2(K_A P_{CO})^{1/2}} \quad (10-11)$$

Taking the inverse of both sides of the equation, then multiplying through by $(P_{CO})^{1/2}$, yields

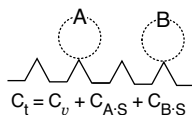
$$\frac{(P_{CO})^{1/2}}{C_{O \cdot S}} = \frac{1}{C_t (K_A)^{1/2}} + \frac{2(P_{CO})^{1/2}}{C_t} \quad (10-12)$$

If dissociative adsorption is the correct model, a plot of $(P_{CO}^{1/2}/C_{O \cdot S})$ versus $P_{CO}^{1/2}$ should be linear with slope $(2/C_t)$.

Dissociative Adsorption



When more than one substance is present, the adsorption isotherm equations are somewhat more complex. The principles are the same, though, and the isotherm equations are easily derived. It is left as an exercise to show that the adsorption isotherm of A in the presence of another adsorbate B is given by the relationship



$$C_{A \cdot S} = \frac{K_A P_A C_t}{1 + K_A P_A + K_B P_B} \quad (10-13)$$

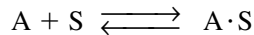
When the adsorption of both A and B are first-order processes, the desorptions are also first order, and both A and B are adsorbed as molecules. The derivations of other Langmuir isotherms are relatively easy.

In obtaining the Langmuir isotherm equations, several aspects of the adsorption system were presupposed in the derivations. The most important of these, and the one that has been subject to the greatest doubt, is that a *uniform* surface is assumed. In other words, any active site has the same attraction for an impinging molecule as does any other active site. Isotherms different from the Langmuir isotherm, such as the Freundlich isotherm, may be derived based on various assumptions concerning the adsorption system, including different types of nonuniform surfaces.

Note assumptions in
the model
and check their
validity.

10.2.4 Surface Reaction

The rate of adsorption of species A onto a solid surface



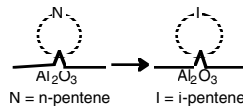
is given by

$$r_{AD} = k_A \left(P_A C_v - \frac{C_{A \cdot S}}{K_A} \right) \quad (10-14)$$

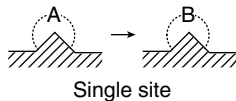
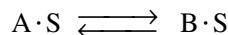
Surface reaction models

After a reactant has been adsorbed onto the surface, i.e., $A \cdot S$, it is capable of reacting in a number of ways to form the reaction product. Three of these ways are:

1. **Single site.** The surface reaction may be a single-site mechanism in which only the site on which the reactant is adsorbed is involved in the reaction. For example, an adsorbed molecule of A may isomerize (or perhaps decompose) directly on the site to which it is attached, such as



The pentane isomerization can be written in generic form as



Each step in the reaction mechanism is elementary, so the surface reaction rate law is

Single Site

$$k_s = \left(\frac{1}{s} \right)$$

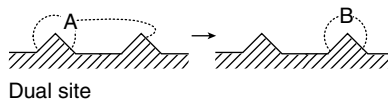
$K_s = (\text{dimensionless})$

$$r_s = k_s C_{A \cdot S} - k_{-s} C_{B \cdot S} = k_s \left(C_{A \cdot S} - \frac{C_{B \cdot S}}{K_s} \right) \quad (10-15)$$

where K_s is the surface-reaction equilibrium constant $K_s = k_s/k_{-s}$

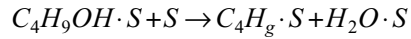
2. **Dual site.** The surface reaction may be a dual-site mechanism in which the adsorbed reactant interacts with another site (either unoccupied or occupied) to form the product.

First type of dual-site mechanism



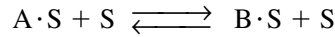
For example, adsorbed A may react with an adjacent vacant site to yield a vacant site and a site on which the product is adsorbed, or as in the case of the dehydration of butanol, the products may adsorb on two adjacent sites.





$$r_s = k_s \left[(C_{C_4H_9OH \cdot S})(C_v) - \frac{(C_{C_4H_g \cdot S})(C_{H_2O \cdot S})}{K_s} \right]$$

For the generic reaction



Dual Site

$$r_s = \left(\frac{\text{mol}}{\text{g-cat} \cdot \text{s}} \right)$$

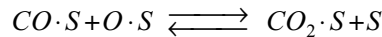
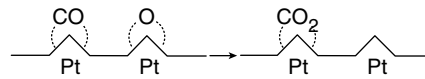
$$k_s = \left(\frac{\text{g-cat}}{\text{mol} \cdot \text{s}} \right)$$

$K_s = (\text{dimensionless})$

the corresponding surface-reaction rate law is

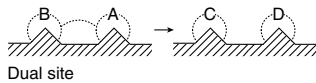
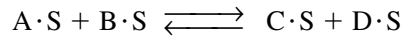
$$r_s = k_s \left(C_{A \cdot S} C_v - \frac{C_{B \cdot S} C_v}{K_s} \right) \quad (10-16)$$

A second dual-site mechanism is the reaction between two adsorbed species, such as the reaction of CO with O.



$$r_s = k_s \left[(C_{CO \cdot S})(C_{O \cdot S}) - \frac{(C_{CO_2 \cdot S})(C_v)}{K_s} \right]$$

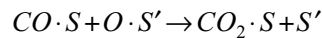
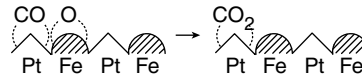
For the generic reaction



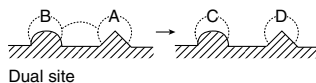
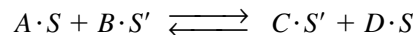
the corresponding surface-reaction rate law is

$$r_s = k_s \left(C_{A \cdot S} C_{B \cdot S} - \frac{C_{C \cdot S} C_{D \cdot S}}{K_s} \right) \quad (10-17)$$

A third dual-site mechanism is the reaction of two species adsorbed on different types of sites S and S', such as the reaction of CO with O.



For the generic reaction



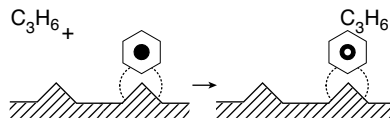
the corresponding surface-reaction rate law is

$$r_s = k_s \left(C_{A \cdot S} C_{B \cdot S'} - \frac{C_{C \cdot S'} C_{D \cdot S}}{K_s} \right) \quad (10-18)$$

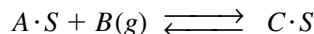
Langmuir–
Hinshelwood
kinetics

Reactions involving either single- or dual-site mechanisms, which were described earlier, are sometimes referred to as following *Langmuir–Hinshelwood kinetics*.

3. **Eley–Rideal.** A third mechanism is the reaction between an adsorbed molecule and a molecule in the gas phase, such as the reaction of propylene and benzene (cf. the reverse reaction in Figure 10-13)



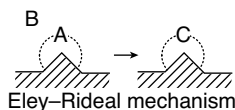
For the generic reaction



the corresponding surface-reaction rate law is

$$r_s = k_s \left(C_{A \cdot S} P_B - \frac{C_{C \cdot S}}{K_s} \right) \quad (10-19)$$

This type of mechanism is referred to as an *Eley–Rideal mechanism*.



$$k_s = \left(\frac{1}{\text{atm} \cdot \text{s}} \right)$$

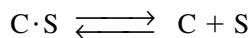
$$K_s = \left(\frac{1}{\text{atm}} \right)$$

10.2.5 Desorption

In each of the preceding cases, the products of the surface reaction adsorbed on the surface are subsequently desorbed into the gas phase. For the desorption of a species (e.g., C)

$$K_{DC} = (\text{atm})$$

$$k_D = \left(\frac{1}{\text{s}} \right)$$



the rate of desorption of C is

$$r_{DC} = k_D \left(C_{C \cdot S} - \frac{P_C C_v}{K_{DC}} \right) \quad (10-20)$$

where K_{DC} is the desorption equilibrium constant with units of atm. Now let's look at the above adsorption from right to left. We note that the desorption step for C is just the reverse of the adsorption step. Consequently, the rate of desorption of C, r_{DC} , is just opposite in sign to the rate of adsorption of C, r_{ADC}

$$r_{DC} = -r_{ADC}$$

In addition, we see that the desorption equilibrium constant K_{DC} is just the reciprocal of the adsorption equilibrium constant for C, K_C

$$K_{DC} = (\text{atm})$$

$$K_C = \left(\frac{1}{\text{atm}} \right)$$

$$K_{DC} = \frac{1}{K_C}$$

in which case the rate of desorption of C can be written

$$r_{DC} = k_D (C_{C \cdot S} - K_C P_C C_v) \quad (10-21)$$

In the material that follows, the form of the equation for the desorption step that we will use to develop our rate laws will be similar to Equation (10-21).

10.2.6 The Rate-Limiting Step

When heterogeneous reactions are carried out at steady state, the rates of each of the three reaction steps in series (adsorption, surface reaction, and desorption) are equal to one another

$$-r'_A = r_{AD} = r_S = r_D$$

However, one particular step in the series is usually found to be *rate-limiting* or *rate-controlling*. That is, if we could make that particular step go faster, the entire reaction would proceed at an accelerated rate. Consider the analogy to the electrical circuit shown in Figure 10-12. A given concentration of reactants is analogous to a given driving force or electromotive force (EMF). The current I (with units of Coulombs/s) is analogous to the rate of

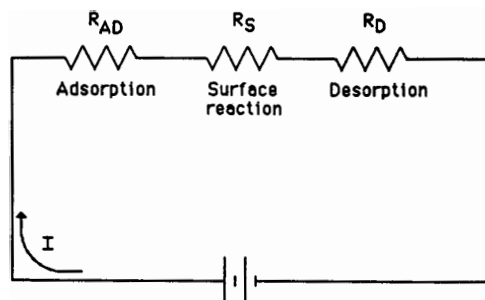


Figure 10-12 Electrical analog to heterogeneous reactions.

reaction, $-r'_A$ (mol/s·g-cat), and a resistance R_i is associated with each step in the series. Because the resistances are in series, the total resistance R_{tot} is just the sum of the individual resistances, for adsorption (R_{AD}), surface reaction (R_S), and desorption (R_D). The current, I , for a given voltage, E , is

$$I = \frac{E}{R_{\text{tot}}} = \frac{E}{R_{AD} + R_S + R_D}$$

The concept of a
rate-limiting step
Who is slowing us
down?

Because we observe only the total resistance, R_{tot} , it is our task to find which resistance is much larger (say, 100 Ω) than the other two resistances (say, 0.1 Ω). Thus, if we could lower the largest resistance, the current I (i.e., $-r'_A$), would be larger for a given voltage, E . Analogously, we want to know which step in the adsorption–reaction–desorption series is limiting the overall rate of reaction.

The approach in determining catalytic and heterogeneous mechanisms is usually termed the *Langmuir–Hinshelwood approach*, since it is derived from ideas proposed by Hinshelwood based on Langmuir’s principles for adsorption.⁷ The Langmuir–Hinshelwood approach was popularized by Hougen and Watson and occasionally includes their names.⁸ It consists of first assuming a sequence of steps in the reaction. In writing this sequence, one must choose among such mechanisms as molecular or atomic adsorption, and single- or dual-site reaction. Next, rate laws are written for the individual steps as shown in the

⁷ C. N. Hinshelwood, *The Kinetics of Chemical Change* (Oxford: Clarendon Press, 1940).

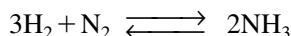
⁸ O. A. Hougen and K. M. Watson, *Ind. Eng. Chem.*, 35, 529 (1943).



preceding section, assuming that all steps are reversible. Finally, a rate-limiting step is postulated, and steps that are not rate-limiting are used to eliminate all coverage-dependent terms. The most questionable assumption in using this technique to obtain a rate law is the hypothesis that the activity of the surface is essentially uniform as far as the various steps in the reaction are concerned.

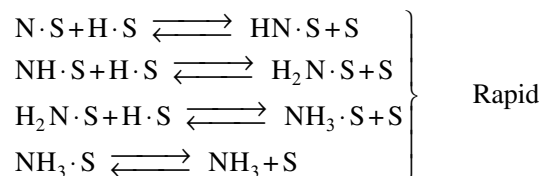
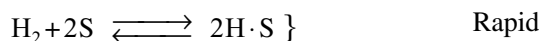
Industrial Example of Adsorption-Limited Reaction

An example of an adsorption-limited reaction is the synthesis of ammonia from hydrogen and nitrogen



over an *iron catalyst* that proceeds by the following mechanism:⁹

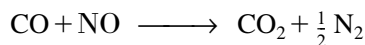
Dissociative
adsorption of
N₂ is rate-limiting



The rate-limiting step is believed to be the adsorption of the N₂ molecule as an N atom.

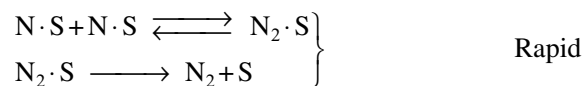
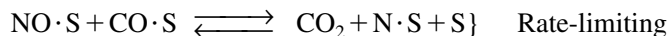
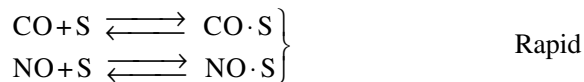
Industrial Example of Surface-Limited Reaction

An example of a surface-limited reaction is the reaction of two noxious automobile exhaust products, CO and NO



carried out *over a copper catalyst* to form environmentally acceptable products, N₂ and CO₂

Surface reaction is
rate-limiting



Analysis of the rate law suggests that CO₂ and N₂ are weakly adsorbed, i.e., have infinitesimally small adsorption constants (see Problem P10-9_B).

⁹ From the literature cited in G. A. Somorjai, *Introduction to Surface Chemistry and Catalysis* (New York: Wiley, 1994), p. 482.